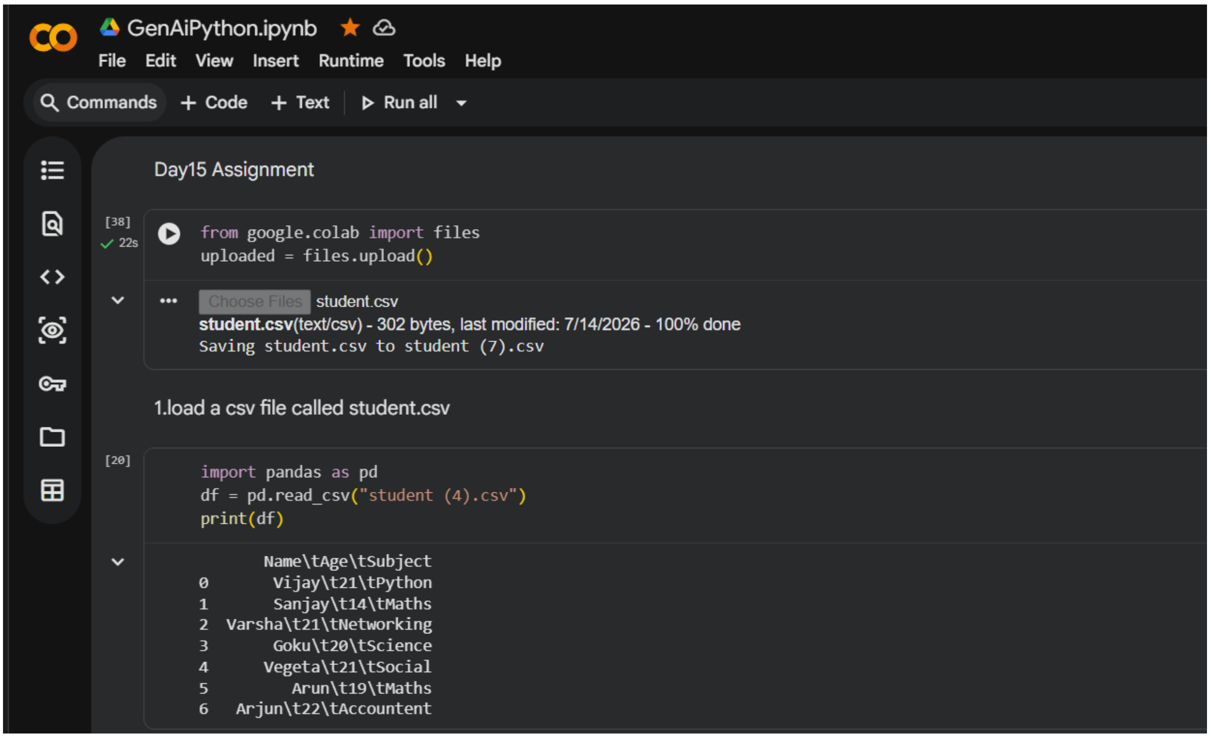
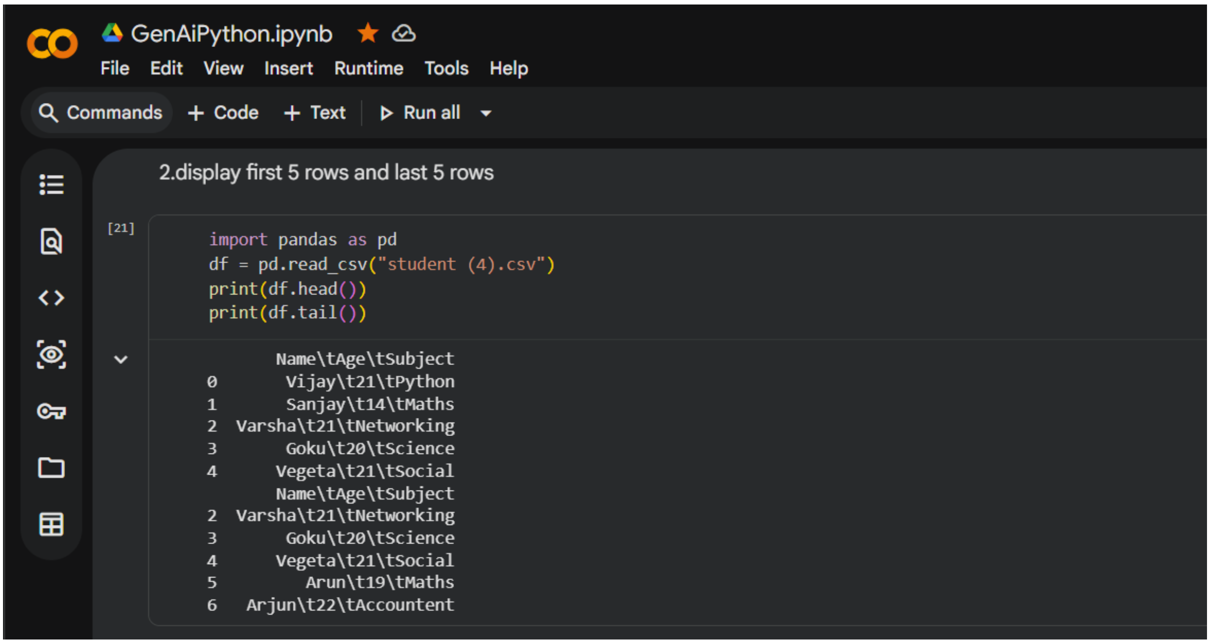
Day 15 Assignment





3.find no of rows and no of columns

[22]
```
import pandas as pd
df = pd.read_csv("student (4).csv")
print(df.shape)
```
```
(7, 1)
```

4.display all columns names

[23]
```
import pandas as pd
df = pd.read_csv("student (4).csv")
print(df.columns)
```
```
Index(['Name\tAge\tSubject'], dtype='object')
```

5.find the datatype of every columns

[24]
```
import pandas as pd
df = pd.read_csv("student (4).csv")
print(df.dtypes)
```
```
Name\tAge\tSubject    object
dtype: object
```

6.generate the statistical summary

[25]
```
import pandas as pd

df = pd.read_csv("student (4).csv")
print(df.describe())
```
```
       Name\tAge\tSubject
count                   7
unique                  7
top     Vijay\t21\tPython
freq                    1
```

7.display three random rows

[26]
```
import pandas as pd
df = pd.read_csv("student (4).csv")
print(df.sample(3))
```
```
      Name\tAge\tSubject
4      Vegeta\t21\tSocial
1       Sanjay\t14\tMaths
2  Varsha\t21\tNetworking
```

8.find how many students belong to each city using value_counts

[32]
```
import pandas as pd
df = pd.read_csv("student (4).csv")
print(df["Subject"].value_counts())
```
```
Subject
Maths         2
Python        1
Networking    1
Science       1
Social        1
Accountent    1
Name: count, dtype: int64
```

9. display only the name column

[34]
```
import pandas as pd
df = pd.read_csv("student (5).csv")
print(df["Name"])
```
```
0     Vijay
1    Sanjay
2    Varsha
3      Goku
4    Vegeta
5      Arun
6     Arjun
Name: Name, dtype: object
```

10.display name and salary columns

```
import pandas as pd
df = pd.read_csv("student (5).csv")
print(df[["Name","Salary"]])
```

```
     Name  Salary
0   Vijay  400000
1  Sanjay  350000
2  Varsha  300000
3    Goku  100000
4  Vegeta   40000
5    Arun  100000
6   Arjun  200000
```

11.display students whose marks are greater than 80

```
import pandas as pd
df = pd.read_csv("student (6).csv")
print(df[df["Marks"]>80])
```

```
     Name  Age     Subject  Salary  Marks
0   Vijay   21      Python  400000     90
1  Sanjay   14       Maths  350000     85
2  Varsha   21  Networking  300000     95
5    Arun   19       Maths  100000     88
```

12.display employee from chennai

```
import pandas as pd
df = pd.read_csv("student (7).csv")
print(df[df["Location"]=="Chennai"])
```

```
     Name  Age     Subject  Salary  Marks Location
0   Vijay   21      Python  400000     90  Chennai
2  Varsha   21  Networking  300000     95  Chennai
```

13.Sort the dataset based on salary in descending order

```
import pandas as pd
df =pd.read_csv("student (7).csv")
print(df.sort_index(ascending=False))
```

```
     Name  Age     Subject  Salary  Marks   Location
6   Arjun   22  Accountent  200000     78  Bangalore
5    Arun   19       Maths  100000     88  Cuddalore
4  Vegeta   21      Social   40000     77  Bangalore
3    Goku   20     Science  100000     60  Bangalore
2  Varsha   21  Networking  300000     95    Chennai
1  Sanjay   14       Maths  350000     85  Cuddalore
0   Vijay   21      Python  400000     90    Chennai
```

14.add a bonus column equal to 15% of salary

```
import pandas as pd
df = pd.read_csv("student (7).csv")
df["Bonus"] = df["Salary"] * 0.15
df.to_csv("student (7).csv", index=False)
print(df)
```

```
     Name  Age     Subject  Salary  Marks   Location    Bonus
0   Vijay   21      Python  400000     90    Chennai  60000.0
1  Sanjay   14       Maths  350000     85  Cuddalore  52500.0
2  Varsha   21  Networking  300000     95    Chennai  45000.0
3    Goku   20     Science  100000     60  Bangalore  15000.0
4  Vegeta   21      Social   40000     77  Bangalore   6000.0
5    Arun   19       Maths  100000     88  Cuddalore  15000.0
6   Arjun   22  Accountent  200000     78  Bangalore  30000.0
```

15.rename the column name to employee_name

```
import pandas as pd
df = pd.read_csv("student (7).csv")
df.rename(columns={"Name":"Employee_name"},inplace=True)
print(df)
```

```
  Employee_name  Age     Subject  Salary  Marks   Location    Bonus
0         Vijay   21      Python  400000     90    Chennai  60000.0
1        Sanjay   14       Maths  350000     85  Cuddalore  52500.0
2        Varsha   21  Networking  300000     95    Chennai  45000.0
3          Goku   20     Science  100000     60  Bangalore  15000.0
4        Vegeta   21      Social   40000     77  Bangalore   6000.0
5          Arun   19       Maths  100000     88  Cuddalore  15000.0
6         Arjun   22  Accountent  200000     78  Bangalore  30000.0
```

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16.delete the bouns column

[66]

```
import pandas as pd
df = pd.read_csv("student (7).csv")
print(df.drop("Bonus",axis=1))
```

```
     Name  Age     Subject  Salary  Marks   Location
0   Vijay   21      Python  400000     90    Chennai
1  Sanjay   14       Maths  350000     85  Cuddalore
2  Varsha   21  Networking  300000     95    Chennai
3    Goku   20     Science  100000     60  Bangalore
4  Vegeta   21      Social   40000     77  Bangalore
5    Arun   19       Maths  100000     88  Cuddalore
6   Arjun   22  Accountent  200000     78  Bangalore
```